

CAPSTONE PROJECT APPROVAL FORM

DEPARTMENT OF COMPUTER ENGINEERING

Semester:

1.0 Project Information

Project Name	Pilgrim Health
Project Area	Computer engineering
Keyword	Hajj central monitoring unit, vital signs monitoring, AI, health band, infrared thermal camera, pilgrim safety.

2.0 TEAM MEMBERS

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503598-3 Capstone Project
Department of Computer Engineering
Taif University

Project Title: Pilgrim Health

TO: Dr. Ahmed Bin Mahfoudh

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9. Rayan Obaidullah Althabeti

Date of submission:

SUBJECT: Submission of Project Proposal for Approval

Enclosed is our group's project proposal. This proposal is submitted for partial fulfillment of the Capstone Project requirement. We understand this proposal, in written report as attached and oral presentation upon scheduled, would undergo a rigorous review process, and we are willing to accept recommendations and modify and resubmit for final approval.

TAIF UNIVERSITY

Fall 2025

CAPSTONE PROJECT PROPOSAL

Pilgrim Health

A Senior Project Submitted To:

Dr. Ahmed Bin Mahfoudh

The College of Computers and Information Technology in

Candidacy for the Degree of

Bachelor in Computer Engineering

by

Osama Saeed AlHarthi

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PROPSAL SUMMARY (ENGLISH)

The graduation project aims to develop an intelligent smart wristband designed to monitor and assist pilgrims during the Hajj season. Every year, millions of pilgrims travel to Saudi Arabia to perform Hajj, creating a need for efficient health and safety management. This project seeks to enhance the well-being of pilgrims through continuous monitoring and real-time analysis of their vital signs. Our contributions focus on integrating a smart health band with an infrared thermal camera to provide real-time health observation, enabling early detection of potential health risks.

Keywords: Hajj, smart wristband, health monitoring, infrared thermal camera, pilgrim safety.

ملخص مقترح البحث عربي

يهدف مشروع التخرج إلى تطوير سوار ذكي متكامل لمراقبة ومساعدة الحجاج أثناء موسم الحج. يشهد موسم الحج سنوياً توافد ملايين الحجاج إلى المملكة العربية السعودية، مما يتطلب نظاماً فعالاً لإدارة السلامة والصحة. يسعى هذا المشروع إلى تعزيز صحة الحجاج من خلال المراقبة المستمرة والتحليل الفوري لمؤشراتهم الحيوية. وتتمثل مساهمتنا في دمج سوار صحي ذكي مع كاميرا حرارية بالأشعة تحت الحمراء لتوفير مراقبة صحية فورية وتمكين الاكتشاف المبكر للمخاطر الصحية المحتملة

Introduction

The Hajj brings together vast numbers of people within compact and crowded locations, creating significant challenges for healthcare and emergency response systems, especially under extreme summer heat. These conditions increase the risk of medical emergencies such as heat exhaustion, low oxygen levels, fainting, and collapse, all of which require quick and accurate medical attention. However, detecting and locating affected individuals in dense crowds is a major obstacle that can delay response and endanger lives.

This project presents a smart and integrated healthcare monitoring system powered by artificial intelligence. The system combines two main technologies: a smart health bracelet that measures vital signs like heart rate and blood oxygen levels while tracking location using GPS, and thermal cameras that identify unusual heat patterns and high-risk areas. The collected data are analyzed by AI algorithms to detect critical cases and send real-time alerts to medical teams with precise locations.

The key contribution of this project is improving the accuracy and speed of emergency response during Hajj, supporting Saudi Arabia's vision of using smart technologies to enhance pilgrims' safety.

Overview and Problem Background:

Millions of pilgrims gather in close proximity during Hajj, creating extremely dense and confined environments., creating major challenges for healthcare systems. The extreme heat and high density increase the risk of health issues such as heat exhaustion, low oxygen levels, fainting, and collapse. The main problem lies in the difficulty of detecting these emergencies quickly and locating the affected individuals accurately within the crowd. Traditional systems often rely on manual reports, causing delays or false alarms that waste time and resources. This project proposes a smart solution that integrates data from a health bracelet and a thermal camera, analyzed by AI algorithms to detect critical cases and improve emergency response efficiency.

Problem Statement:

Millions of pilgrims must endure scorching heat and overcrowding easily leading to serious medical conditions such as heat exhaustion, dehydration, or collapse during Hajj season. It is extremely difficult to identify these medical emergencies in real time due to the large crowds and extensive areas. Current monitoring solutions merely depend on personal observation or basic wearable technologies, which are not sufficient to identify emergencies in a timely manner or with accurate precision.

Traditional systems lack intelligent analysis and automatic detection, and thus it is difficult for medical teams to respond in time. Delayed intervention can cause severe health complications or even mortality. Furthermore, existing wearable technologies track only limited individual information without aggregating it to a centralized intelligent system with the ability of scale health risk analysis and prediction.

This project suggests an artificial intelligence-based smart and integrated healthcare monitoring system. Technology combines two technologies of prime importance: a smart bracelet that tracks vital parameters such as heart rate and blood oxygen saturation while tracking the pilgrim's location through GPS, and infrared thermal imaging cameras to detect abnormal thermal patterns and mark high-risk areas. These data are then processed in real time using AI algorithms to recognize severe health cases and send instant notifications with precise location information to medical personnel for immediate response.

The intelligent strategy aims to minimize the response time, prevent severe incidents, and enhance overall health and safety of pilgrims during Hajj, aligned with Saudi Arabia's Vision goals to adopt innovative technologies to improve public health and intelligent crowd management.

Literature Review

Health risks during Hajj rise noticeably due to factors such as extreme heat, intense physical effort, and the sheer density of the crowd. Many sources highlight that early detection of warning signs like heat exhaustion or a sudden drop in oxygen levels can dramatically reduce the chances of serious medical incidents, especially when medical teams cannot immediately reach every individual.

Wearable health devices, particularly smart wristband, have shown strong potential in continuous vital sign monitoring. Unlike manual health checks, these devices track data in real time, making them far more suitable for large-scale gatherings where direct medical supervision is limited.

The use of infrared thermal imaging has also proven to be an effective method for detecting abnormal temperature changes in crowded environments. It has already been adopted in high-traffic areas such as airports, which makes it a strong candidate for application during Hajj to identify signs of heat stress or fever at an early stage.

When it comes to processing health information, AI-driven analysis makes a significant difference. Instead of simply collecting raw readings, intelligent systems can interpret the data, detect patterns, and notify medical teams only when a situation is genuinely critical. This targeted approach helps reduce unnecessary alerts and improves response speed.

There is also strong emphasis on connecting multiple health data sources into one unified system. When thermal readings are combined with vital sign data from wearables, detection accuracy increases, and false alarms are minimized, resulting in a more reliable and efficient monitoring setup.

Objectives

In this project, we aim to achieve the following objectives:

- 1- Monitoring pilgrims' health in real time using a smart bracelet that measures vital signs such as heart rate and oxygen levels.
- 2- Early detection of emergency cases like low oxygen levels, heat stress, or abnormal heart rate.
- 3- Identifying high-risk or overcrowded areas through thermal cameras and heat pattern analysis.
- 4- Helping medical teams respond faster to emergencies by providing accurate location information.
- 5- Reducing pressure on medical staff by minimizing false alerts and improving efficiency.
- 6- Creating a safer and healthier environment for all pilgrims using smart and modern technology.

Proposed Methodology

The project will follow a V-Model methodology, a structured approach that emphasizes verification and validation at each phase to ensure a reliable and robust system. This method ensures that the final integrated system addresses the high-level security and health requirements of the Hajj environment.

System Design Model

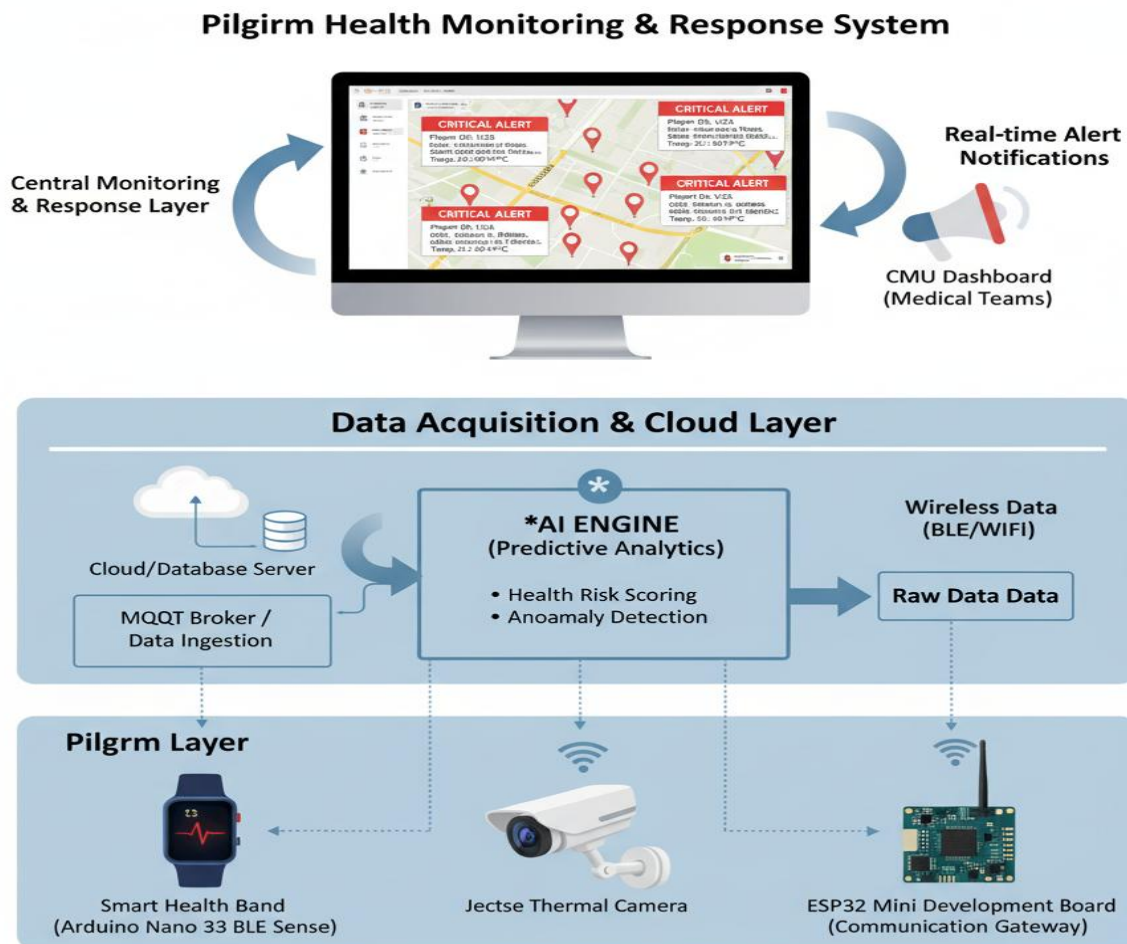
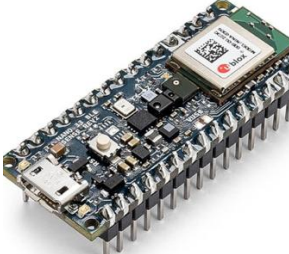


Figure 1 System Model Diagram

Tools & Software

Hardware components

Table 1: Hardware components

Component	Description	Image
Arduino uno	The Arduino Uno is an open-source microcontroller development board based on the ATmega328P microcontroller. It provides a simple and accessible platform for prototyping electronic projects, using both hardware and software.	
ARDUINO NANO 33 SENSE REV2	The Arduino Nano 33 BLE Sense Rev2 is a compact, sensor-rich, Bluetooth Low Energy-enabled board, designed for IoT, embedded intelligence, and machine learning (TinyML) applications.	
Jectse Thermal	A Thermal Imaging Camera , also known as an Infrared (IR) Camera , is a device that detects infrared radiation (heat) emitted by objects and converts it into a visual image .	

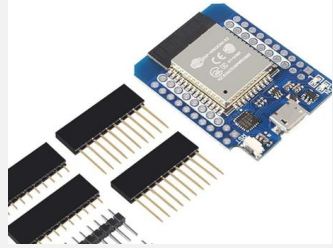
Lithium Polymer Ion Battery 3.7V 500mAh	is a compact, lightweight, and rechargeable power source ideal for small electronic devices and DIY projects. It delivers a stable 3.7-volt output with a capacity of 500 milliamp-hours, providing reliable energy for portable systems such as Arduino projects, wearables, sensors, and IoT applications	
Arduino Wires	flexible, insulated electrical wires used to connect electronic components to Arduino boards and other microcontrollers.	
GPS Sensor	GPS (Global Positioning System) sensor, which enables it to determine its precise location. This information is crucial for accurate navigation, mapping the disaster area, and coordinating with other rescue teams	
ESP32 Mini Development Board (optional)	Is a compact, low-power, and versatile microcontroller board designed for IoT, wireless communication, and embedded projects . It features a dual-core processor , built-in Wi-Fi and Bluetooth (BLE) , and multiple GPIO pins for connecting sensors, actuators, and other electronic modules.	

Table 2: costs of components

Component	Cost
Arduino UNO	23.04\$
GPS Sensor	12\$
ESP32 Mini Development Board	19\$
Lithium Polymer Ion Battery 3.7V 500mAh	8.18\$
ARDUINO NANO 33 SENSE REV2	70\$
Jectse Thermal	45\$
Total cost	177.22\$

Software components

Table 3: Software components

Application	Programming Language	Description	Use for
Arduino IDE	C/C++	Node MCU can be Programmed using C++ with an addition of special methods and functions, C++ is a human-readable programming language and using Arduino IDE it can create a 'sketch' (the name given to code files), which is processed and compiled to machine language	Arduino Uno/ ARDUINO NANO 33 SENSE REV2
Arduino IDE/ ESP-IDF	C/C++	supported by a wide range of software development environments and tools , allowing users to program, debug, and deploy IoT and embedded applications easily.	ESP32Mini Development Board

Timetable or Schedule

Task	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13	W14	W15	Hours (H)	%
Project Proposal Finalization	12	12	8	8	8	8										56	18%
Detailed Requirements Gathering & Specification			4	4	4	4										16	5%
Study of Current Technology/Literature Review			4	4	4	4										16	5%
System High-Level Design (Architecture & Layers)							16	16	12	12	12					68	22%
Hardware & Sensor Selection/Verification							4	4	4	4	12	12				40	13%
Project Management/Weekly Meetings	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	60	20%
Presentation Preparation & Final Report Outline												8	8	8	24	48	16%
TOTAL HOURS PER WEEK	16	16	20	20	20	20	24	24	20	20	28	24	12	12	28	304	

Semester 1 Schedule

Figure 2 Activity Plan Part I

Task	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13	W14	W15	Hours (H)	%
Hardware Prototyping & Individual Component Coding	16	16	16	16	8	8	8	4								92	30%
System Integration & Data Flow Testing (BLE/MQTT/Cloud)				8	8	8	8									32	11%
AI Engine Development (Model Training & Predictive Analytics)							4	16	16	8	8					52	17%
Final System Validation & Testing (Anomaly Detection)										8	8	4	4			24	8%
Final Documentation (Technical Report)												4	8	8		20	7%
Presentation and Poster Preparation												4	4	8	8	24	8%
Project Management/Weekly Meetings	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	60	20%
TOTAL HOURS PER WEEK	20	20	20	28	20	20	24	24	20	20	20	16	20	20	12	304	

Semester 2 Schedule

Figure 3 Activity Plan Part II

Expected Benefits

By the end of this project, we expect to obtain the following benefits:

The development of a smart bracelet to monitor pilgrims during the Hajj season offers several benefits at the individual, societal, and organizational levels, including the following:

1. Enhancing pilgrims' safety: It contributes to monitoring vital signs and body temperature, and early detection of signs of heat stress, dehydration, and other health problems requiring urgent medical intervention.
2. Providing immediate health information: By integrating an infrared thermal camera, pilgrims' health is monitored, improving the accuracy and speed of detecting emergency cases among pilgrims.
3. Improving healthcare management during Hajj: By collecting and analyzing pilgrim data, the system helps identify high-risk cases and prioritize them to increase response efficiency.
4. Reducing emergency cases: Early detection and immediate alerts contribute to reducing critical health referrals during the Hajj season.
5. Supporting data-driven decision-making: Health data can be used to improve future planning for developing healthcare services during the Hajj season and assisting relevant authorities in crowd management and safety.
6. Scalability and Applicability: The fences can be expanded in the future to include Umrah services or public events, making them a viable and reusable solution for public health monitoring.

References

1. [Yahuza Bello & Emanuel Figetakis, "IoT-based Wearables: A comprehensive Survey."](#)
 2. [Alamsyah, Mery Subito, Ikhlayel Mohammad, Eko Setijadi & others, "Internet of things–based vital sign monitoring system." *Int. J. Electrical & Computer Eng.*, Vol.10 No.6, 2020.](#)
 3. [J. W. Chin, et al, An Infrared Thermography Model Enabling Remote Body Temperature Screening Up to 10 Meters. \(2021\)](#)
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